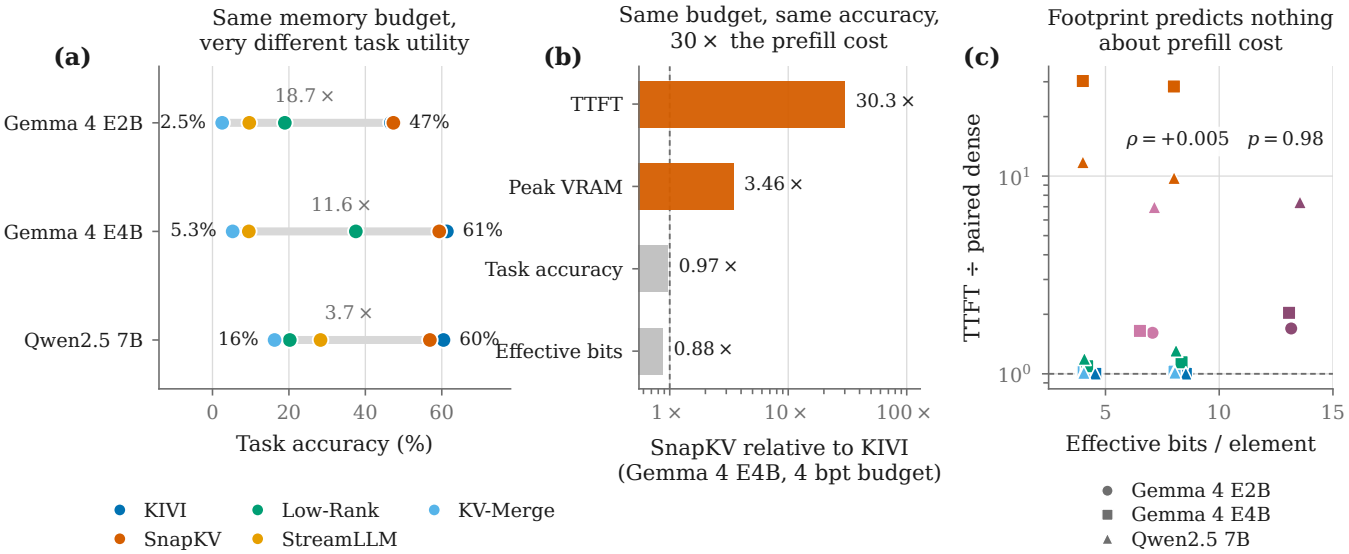
(a)
Same memory budget,
very different task utility
Gemma 4 E2B
Gemma 4 E4B
Qwen2.5 7B
18.7 ×
11.6 ×
3.7 ×
2.5%
47%
5.3%
61%
16%
60%
0
20
40
60
Task accuracy (%)
(b)
Same budget, same accuracy,
30 × the prefill cost
TTFT
Peak VRAM
Task accuracy
Effective bits
30.3 ×
3.46 ×
0.97 ×
0.88 ×
1 ×
10 ×
100 ×
SnapKV relative to KIVI
(Gemma 4 E4B, 4 bpt budget)
(c)
Footprint predicts nothing
about prefill cost
ρ = +0.005 p = 0.98
TTFT ÷ paired dense
10^1
10^0
5
10
15
Effective bits / element
Gemma 4 E2B
Gemma 4 E4B
Qwen2.5 7B
KIVI
SnapKV
Low-Rank
StreamLLM
KV-Merge